Small L^2 Error Is Not Enough: Function-Space Consistency for Neural Operators

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Abstract

Neural operators are an intensely studied class of machine-learning surrogates for the maps from parameters to solutions of partial differential equations. The prevailing supervised recipe is pointwise regression: one samples the ground-truth solution at a fixed set of nodes and then fits a network so that its pointwise outputs match those samples in a relative L^2 sense. We argue that this paradigm, although effective at producing predictions that look correct and report small relative L^2 error, is structurally insufficient for scientific use. Pointwise labels discard the function-space information that defines the solution as a continuous object, so the learned operator carries no guarantee about values away from the training nodes or about derived quantities obtained by postprocessing, such as gradients, divergences, and integrals. The failure is sharpest for fields that live in function spaces such as $H(\mathrm{div})$, whose conforming discretization by Raviart-Thomas or Brezzi-Douglas-Marini elements enforces continuity of the normal flux across element interfaces rather than interpolating point values. On a nonlinear diffusion-reaction benchmark, a mainstream pointwise neural operator matches and even slightly beats a structure-aware $H(\mathrm{div})$-conforming neural operator in relative L^2 error, yet is far worse in divergence, conservation residual, and boundary-flux postprocessing; a representation floor shows that this failure is set by the output representation rather than by training. We close with a research agenda for function-space-consistent operator learning.

CCS Concepts

• **Mathematics of computing** → **Partial differential equations**; *Mathematical software*; • **Computing methodologies** → *Machine learning*.

Keywords

neural operators, physics-informed machine learning, finite elements, function spaces, a posteriori certification, postprocessing

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1 Introduction

A neural operator approximates the solution map $p \mapsto u$ of a parametric partial differential equation, where p is a coefficient, geometry, or forcing field and u is the corresponding solution [6, 7, 9]. The mainstream training recipe is remarkably uniform across architectures. One generates a dataset of solutions with a high-fidelity numerical solver, evaluates each solution *pointwise* on a fixed grid of nodes, and fits the network so that its pointwise predictions match these labels, almost always under a relative L^2 loss. Success is then declared when the held-out relative L^2 error is small and the predicted fields look like the reference fields.

This paper takes the position that pointwise supervised regression, despite its empirical popularity, is often an incomplete abstraction for scientific computing, and that closing the resulting gap is an important challenge for the field. Our central claim is simple to state. A small pointwise error on a fixed node set says almost nothing about the two properties that scientific workflows actually need: *off-node fidelity*, accuracy at points never seen during training, and *postprocessing consistency*, accuracy of derived quantities obtained by applying differential and integral operators to the predicted field. These derived quantities, including fluxes, conservation balances, energies, reaction rates, are not edge cases but the normal output of scientific computing, from uncertainty quantification and inverse problems to design optimization and multiphysics coupling. A neural operator queried only at fixed nodes can reproduce the evaluations closely while the quantities computed from them by postprocessing turn out far less accurate, or even qualitatively wrong, making it unusable for precisely the tasks that motivate building a fast surrogate.

The reason pointwise supervision is fragile is that a finite set of pointwise evaluations does not determine a function. Reconstructing a continuous field from those evaluations requires interpolation information, that is, a choice of basis and of which function space the field belongs to. Standard pointwise training passes none of this information to the network. The labels are bare numbers at bare coordinates, stripped of the structure that makes them the trace of a genuine solution. As a result, the operator is free to interpolate between nodes in any way that keeps the nodal residual small, including ways that oscillate, violate conservation, or blow up under differentiation.

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Pointwise learning *may* fail rather than *must* fail. When the solution manifold is well represented by piecewise-linear polynomials, the implicit interpolation is benign and off-node prediction can be quite good. The danger appears when the field lives in a function space whose conforming representation is not built from point values. The flux σ of a diffusion problem lives in $H(\mathrm{div})$, whose conforming discretization uses Raviart-Thomas or Brezzi-Douglas-Marini elements, whose degrees of freedom are integral moments of the normal component on element edges and which therefore enforce normal-component continuity across interfaces rather than interpolating point values [1, 3, 12]. For such fields, pointwise supervision is poorly matched to the field, and the error surfaces once one takes a divergence or integrates a normal trace.

A second, more practical, manifestation concerns numerical integration. Many downstream quantities are integrals, evaluated at non-uniform quadrature nodes rather than on the uniform grid used for training, so an operator accurate on the grid is required to generalize to the quadrature points with no control on the resulting error. We treat this grid-to-quadrature shift as a genuine source of generalization error and return to it in the research agenda.

This position relates to, but differs from, an active line of structure-preserving and physics-informed operator learning. Divergence-free and conservation-law architectures hard-wire specific invariants into the network [4, 13], and physics-informed neural operators add equation residuals to an otherwise pointwise training loss [8]. We do not propose another such architecture. Our contribution is a diagnosis: a single aggregate relative L^2 score systematically hides postprocessing error; a reconstruction baseline shows the cause lies in the output representation rather than in training; and a variationally correct residual supplies a ground-truth-free certificate of the derived quantities that scientific workflows actually consume.

2 Why Pointwise Supervision Is Insufficient

A function is not just its point values. Let u be sampled for training at spatial points $\{x_i\}$, so that supervision constrains only the values $\{u(x_i)\}$. Two functions that agree at every x_i can still disagree arbitrarily between them and have wildly different derivatives. Recovering a function from finitely many point values is well posed only once we commit to an interpolation rule and a function space, a commitment the finite element method makes explicit through its choice of space and degrees of freedom [5]. Pointwise neural operator training leaves it implicit, so the network supplies its own interpolation, tuned to minimize the mismatch at the nodes rather than to respect the numerical solver that generated the data.

Postprocessing amplifies what L^2 hides. Relative L^2 error averages the discrepancy and is forgiving of small, high-frequency, structured errors. Differential postprocessing is not. If the prediction error is a small oscillatory field e, then $\|e\|_{L^2}$ can be tiny while $\|\nabla \cdot e\|_{L^2}$ is large, because differentiation multiplies each Fourier mode by its frequency. A flux prediction can therefore be visually perfect and L^2-accurate yet produce a divergence, a boundary flux, or a conservation residual that is badly wrong. Section 3 makes this quantitative on a concrete benchmark.

Grid versus quadrature. Training data are conventionally placed on uniform grids, partly for convenience of convolution. Integrals of interest, however, are at best estimated with quadrature rules whose nodes are typically non-uniform. A neural operator that has only ever been supervised on a uniform grid has no reason to be accurate at quadrature points, and the integration error inherits this uncontrolled off-node behavior. The community currently has no standard way to measure or bound this grid-to-quadrature generalization gap.

The opportunity. One remedy is to put the discarded structure back into the learning problem. Instead of fitting bare point values, one can require the predicted output to lie in the solution's natural function space and train it to minimize the variationally correct residual of the equation [11]—the same residual whose minimizer defines the true solution. The loss then controls the norm in which the intended postprocessing is stable, so that the derived quantities the application cares about, and not only the point values, are well defined. The residual of the governing equation moreover serves as an a posteriori certificate that can be checked at inference time without the ground truth. The next section makes this concrete on a nonlinear benchmark by contrasting a pointwise neural operator with a structure-aware one.

3 A Concrete Illustration: Nonlinear Diffusion-Reaction Problem

Setup and two operators. We consider the first-order system formulation of a nonlinear diffusion–reaction problem on $\Omega = (0,1)^2$,

$$\sigma - p_1 \nabla u = 0, \qquad -\nabla \cdot \sigma + p_2 u^3 - p_3 = 0, \tag{1}$$

with $u = 0$ on $\partial\Omega$. The random parameter $p = (p_1, p_2, p_3)$ consists of three spatially varying fields. The diffusion and reaction coefficients are given by $p_1 = \exp(m_1)$, $p_2 = \exp(m_2)$, where m_1 and m_2 are independent samples drawn from Gaussian random fields with Matérn covariance. The forcing field is represented as a superposition of Gaussian bumps,

$$p_3 = \sum_{i=1}^{49} z_i \chi_i(x), \qquad \chi_i(x) = \frac{1}{(\ell\sqrt{2\pi})^2} \exp\left(-\frac{|x - c_i|^2}{2\ell^2}\right), \tag{2}$$

where the centers $c_i = (m/8, n/8)$, $m, n = 1, \ldots, 7$, form a 7×7 grid, the width is $\ell = 0.08$, and the coefficients z_i are independently sampled from the uniform distribution $\mathcal{U}(-1, 1)$.

We assume the finite element discretization uses a 128×128 triangular mesh with u in CG_2 space (continuous Lagrange element of degree 2) and, crucially, the flux σ in RT_1 (Raviart-Thomas element of order 1, which conforms to $H(\mathrm{div})$). We compare two neural operators on identical held-out parameters.[1] The first is a mainstream *pointwise* model: the Fourier neural operator [7] that regresses the pointwise evaluations (σ, u) on a uniform 129×129 grid under a relative L^2 loss. Because this model outputs only point values, we evaluate it by reconstructing its output as a piecewise-linear $\mathrm{CG}_1 \times \mathrm{CG}_1$ field and interpolating that into the same $\mathrm{RT}_1 \times \mathrm{CG}_2$ space as the reference, so that the computation of every error, residual, and integral are well aligned for both neural operators. The second is a *structure-aware* model, the reduced basis neural operator of Qiu et al. [11], following the variationally correct residual

[1] Both operators have comparable parameter counts and are trained and evaluated on samples from the same distribution; code is publicly available at https://github.com/qy849/nonlinear_vcno.

Table 1: Pointwise versus structure-aware operator learning on identical held-out samples. Entries are mean relative errors, except the conservation residual $\|-\nabla\cdot\sigma + p_2u^3 - p_3\|^2_{L^2}$. The pointwise model matches the structure-aware one in L^2 but is far worse on every quantity that touches the flux.

	Pointwise FNO	Reconstruction baseline	Structure aware
Rel err $L^2 \times L^2$	3.4×10^{-2}	7.8×10^{-4}	4.5×10^{-2}
Rel err $H(\mathrm{div}) \times H^1$	1.1×10^{-1}	3.5×10^{-2}	3.5×10^{-2}
Conservation residual	2.2	2.4×10^{-1}	4.1×10^{-2}
Q_1 reaction integral	8.9×10^{-2}	8.8×10^{-4}	8.0×10^{-2}
Q_2 right-edge flux	1.9×10^{-1}	1.5×10^{-4}	8.6×10^{-2}

regression framework of Bachmayr et al. [2]. It uses a hybrid representation: the finite element basis functions are predetermined and fixed, and the network predicts only the coefficients that multiply them, so every field it produces is, by construction, a member of the $H(\mathrm{div}) \times H^1$-conforming finite element space. To keep the prediction low dimensional, the full degrees of freedom are reduced by proper orthogonal decomposition to a 512-mode basis in the $H(\mathrm{div}) \times H^1$ norm, and the network predicts these reduced coefficients rather than the full degrees of freedom. It is trained against a first-order system least-squares residual

$$\mathcal{J}(\sigma, u; p) = \|\sigma - p_1\nabla u\|^2_{L^2} + \|-\nabla\cdot\sigma + p_2u^3 - p_3\|^2_{L^2}, \qquad (3)$$

that is provably equivalent to the solution error in the norm induced by the equation, globally in the linear case [2], and locally in a neighborhood of the stable solution branch, for the nonlinear residual considered in [10], so the residual it minimizes also serves, at inference and without ground truth, as an a posteriori estimate of the prediction error. To isolate the error introduced by the representation from that of the neural operators, we also report a *reconstruction baseline*: the error obtained by applying the same reconstruction used for the FNO prediction to the exact solution evaluated on the grid. We stress that the two neural operators differ in architecture, loss, and output representation at once, so the head-to-head comparison alone cannot attribute the gap to any single factor. It is the reconstruction baseline that isolates the representation, since it removes both the network and the loss and exposes what pointwise supervision can achieve in principle.

Comparable L^2 error, far worse physical accuracy. Table 1 reports both neural operators and the reconstruction baseline on the same test samples. The pointwise model is not worse on the metric it optimizes: its mean relative L^2 error is in fact slightly *better* than the structure-aware model. However, its relative $H(\mathrm{div}) \times H^1$ error is three times larger, and its conservation residual is roughly fifty times larger. Figure 1 shows that the predicted flux is visually almost indistinguishable from the reference. Figure 2 visualizes the corresponding divergence field and its error, computed either by finite differences on the grid or by differentiating an $H(\mathrm{div})$ finite element reconstruction. Although the predicted divergence reproduces the large-scale features of the reference, the error exhibits fine-scale spatial structures that are not evident from the aggregate error metrics alone.

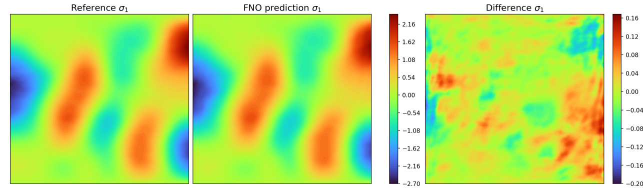

Figure 1: Flux component $\sigma_1 \in H(\mathrm{div})$ for a held-out sample: reference (left), pointwise FNO prediction (center), and their difference (right). The prediction is visually indistinguishable from the truth and has small relative L^2 error.

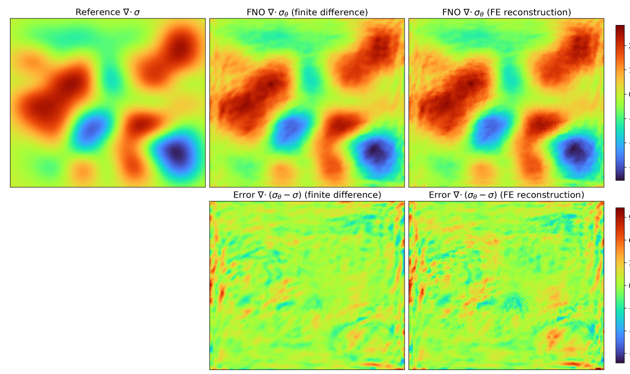

Figure 2: Top: for the held-out sample of Figure 1, the reference divergence $\nabla\cdot\sigma$ (left) is smooth, while the divergence of the pointwise FNO flux is contaminated by high-frequency error, whether computed by a finite difference of the predicted grid (center) or by differentiating an $H(\mathrm{div})$ reconstruction (right). Bottom: the corresponding divergence errors $\nabla\cdot(\sigma_\theta - \sigma)$, which are similar for the two methods.

Postprocessing separates the two. We extract two downstream quantities of interest a practitioner might need,

$$Q_1 = \int_{\Omega_c} p_2u^3\,dx, \qquad Q_2 = \int_\Gamma \sigma\cdot n\,ds, \qquad (4)$$

a reaction integral over a central box $\Omega_c \subset \Omega$ and the outward flux through the right edge of the boundary $\Gamma = \{1\} \times (0, 1)$. The quantity Q_1 is obtained from the state u and the errors on it are comparable. The quantity Q_2 is the normal-trace postprocessing of the flux, exactly where $H(\mathrm{div})$ structure matters, and here the pointwise model is more than twice as inaccurate.

The cause is representational. The reconstruction baseline isolates the error introduced by the pointwise representation. Evaluating the *exact* solution on the grid and reconstructing it in the finite element spaces gives a negligible relative $L^2 \times L^2$ error of 7.8×10^{-4}, yet a relative $H(\mathrm{div}) \times H^1$ error of 3.5×10^{-2} and a fairly large conservation residual of 2.4×10^{-1}. Thus, even exact point values do not preserve the derivative and conservation structure of the solution. This effect is amplified for the FNO: despite an $L^2 \times L^2$ error comparable to that of the structure-aware model, its $H(\mathrm{div}) \times H^1$ error is more than three times larger and its conservation residual is over an order of magnitude larger. The same pattern appears in the quantities of interest: the two models have similar errors for the reaction integral Q_1, whereas the flux-sensitive Q_2 is substantially more accurate with the structure-aware model. In contrast, the structure-aware neural operator predicts directly in the $H(\mathrm{div}) \times H^1$-conforming space and therefore avoids this representational mismatch. Moreover, the residual loss (3) used in training can also be accurately evaluated at inference stage. Consequently, the structure-aware neural operator not only preserves the function-space structure required by the PDE, but also provides a computable a posteriori indicator of its prediction accuracy.

4 Where Should We Go Next?

The diffusion-reaction case is a single instance of a general phenomenon. We propose the following research directions for physics-informed machine learning.

1. Output spaces with the right structure. Operator architectures should predict directly into conforming finite element spaces appropriate for the underlying PDE solutions. For example, fluxes should be represented in $H(\mathrm{div})$-conforming spaces, while $H(\mathrm{curl})$-conforming vector fields, such as those arising in electromagnetics, should be represented in $H(\mathrm{curl})$-conforming spaces rather than via Lagrange interpolation. The prediction targets should therefore be the associated finite element degrees of freedom rather than point values [1, 3]. The reduced basis neural operator we show is one concrete realization of this idea, and its strengths are also its limitations. Its conforming Raviart-Thomas basis supplies exactly the interpolation information the pointwise model lacked, but it is chosen by hand. Designing such an element for every field and equation is laborious, and whether a learned element would serve better is open. In addition, its precomputed linear reduced basis, from proper orthogonal decomposition, shrinks the search space to a neighborhood of the solution manifold, but linear reduced bases are well known to be ineffective for transport-dominated problems with slowly decaying Kolmogorov width. The reduction also appears important in practice, as we found it difficult to learn the finite element degrees of freedom directly, through either supervised or unsupervised training, without first reducing the output dimension. The central challenge, therefore, is to make such structure-preserving output layers more expressive and automatic—for instance, through decoders that faithfully represent the solution without restricting it to a precomputed linear subspace, while making them as easy to use as a dense regression head.

2. Losses that control postprocessing norms. Training should penalize the error in the norm under which the intended postprocessing is stable, and should incorporate the variationally correct residual of the governing equation as supervision rather than as a diagnostic that is applied only after the fact. When that residual is a first-order least-squares functional equivalent to the solution error in the relevant norm (globally for linear problems and locally near a stable solution branch for nonlinear problems [2, 10]), minimizing it during training directly controls the solution error [11]. The open problem is to identify, for a given downstream operator, the weakest norm that still guarantees a small postprocessed error, and then to design training procedures that learn in that norm efficiently.

3. Certified inference. The equation residual is a ground-truth-free a posteriori error indicator, and operators should ship with such certificates so that an individual prediction can be accepted, flagged for review, or corrected at test time [11]. The remaining challenge is to turn these residual estimates into rigorous and quantity-of-interest-specific error bounds rather than into calibrated heuristics that hold only on average.

4. Benchmarks that measure what matters. Neural operators should be judged on the quantities that scientific workflows actually consume rather than on a single aggregate score that might hide them. We call for benchmarks that report derived-quantity error, conservation residuals, and off-node error alongside the usual relative L^2 error, and that evaluate neural operators at the non-uniform quadrature nodes and random off-grid points where integrals and functionals are actually computed, not only on the uniform training grid. Training schemes such as randomized or quadrature-matched node sampling, which target accuracy at those locations, are a natural companion.

What we are really asking for. We do not argue for this particular architecture. The lesson is more general: once enough prior knowledge is injected, through the interpolation information that determines the function space inhabited by the output and a reduced representation that constrains the search to physically realizable solutions, a structure-aware neural operator outperforms a pointwise Fourier neural operator on the quantities that downstream tasks in scientific workflows depend upon. The challenge is to obtain this advantage with far less human intervention and far greater generality, so that the appropriate structure is discovered or adapted automatically across very different equations rather than hand-built anew for each one.

5 Conclusion

Pointwise supervised regression made neural operators easy to train and easy to score, but the score that it optimizes is not the score that science needs. A small relative L^2 error on a fixed grid certifies neither off-node fidelity nor postprocessing consistency, and for fields whose natural home is a function space such as $H(\mathrm{div})$, the gap between a prediction that looks correct and a prediction that is actually usable can be large. Our diffusion-reaction study makes that gap concrete, and it shows that the missing ingredient is precisely the function-space and variational structure that pointwise labels discard. A structure-aware neural operator that restores this structure, through a conforming output representation together with a residual loss that is equivalent to the solution error, already recovers the accuracy that the pointwise model loses. No single such construction is the final answer. The broader lesson extends beyond any particular architecture: restoring function-space structure closes the gap between predictions that look correct and predictions that are actually usable.

Ethics and Privacy Statement

This work uses only synthetic, solver-generated data and involves no human subjects or personal data. Its aim is to reduce the risk of silently inaccurate predictions in scientific and engineering pipelines.

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